Could I recover the application after deleting an order?

I built a small orders application on AWS, automated its backups and tested recovery on a separate server.

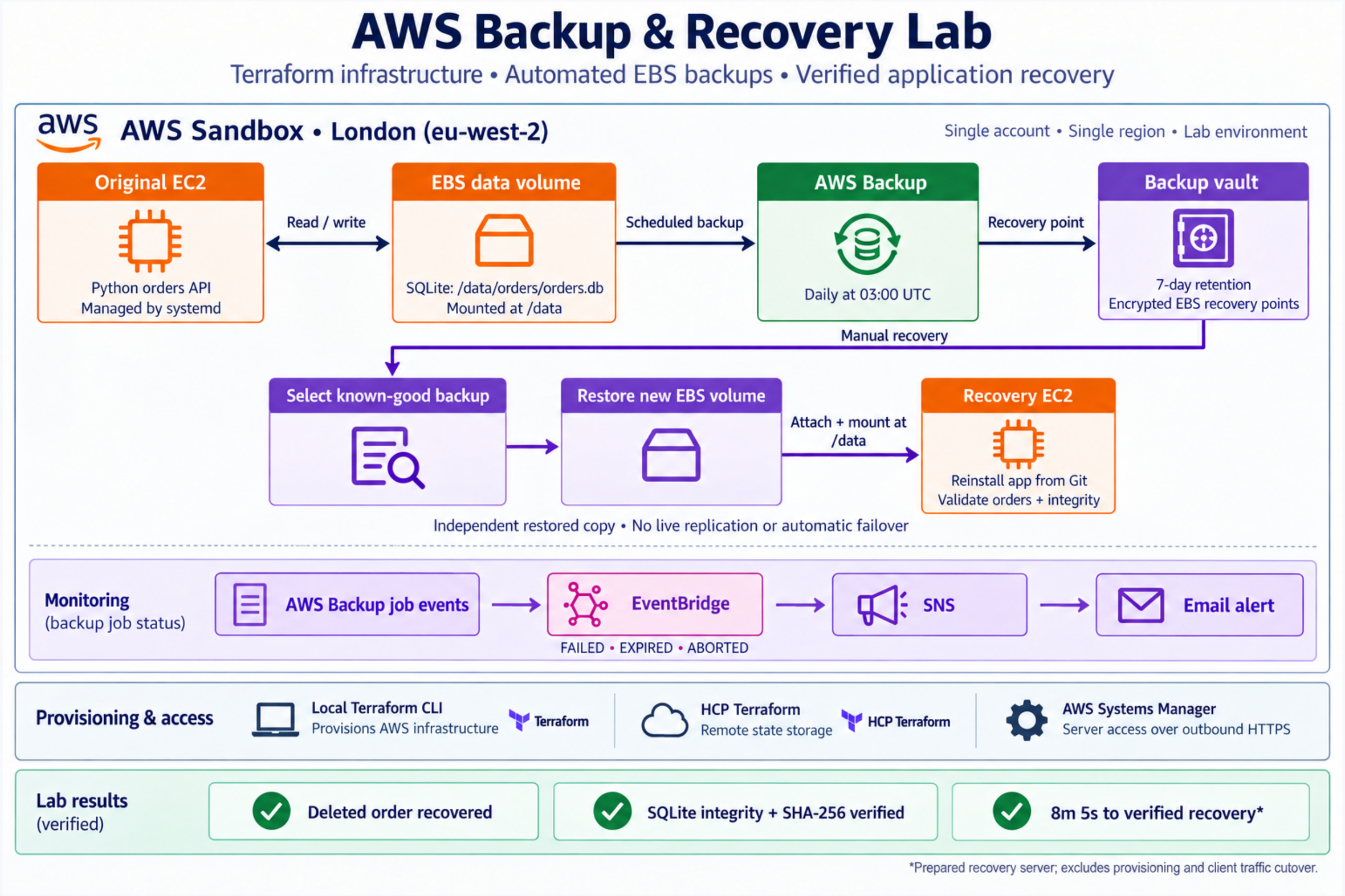
8 minutes 5 seconds

From the recorded deletion to verified application recovery.

The recovery server was already prepared. This timing excludes server provisioning and client traffic switching.

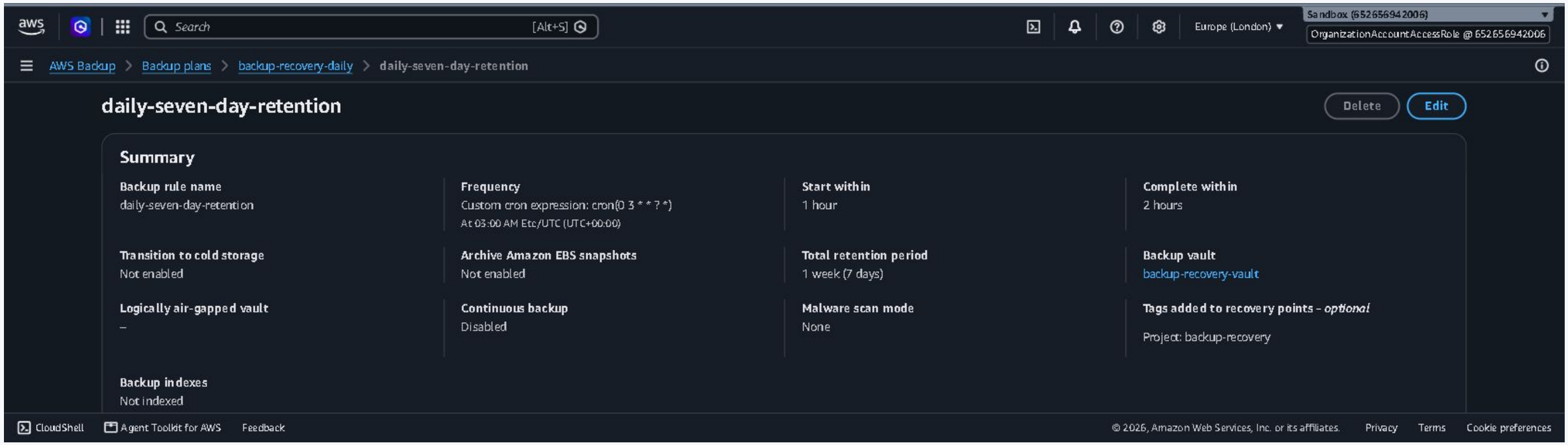
A walkthrough of the project and the evidence collected on 12 to 14 October 2026.

How the project connects



The backup schedule

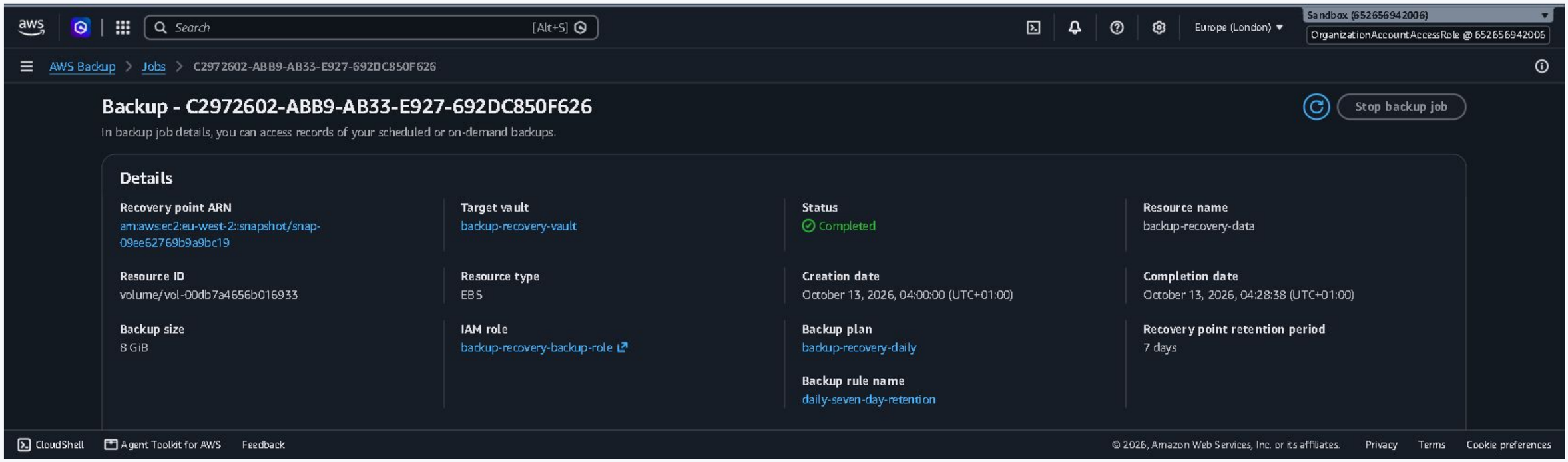
AWS Backup runs the plan daily at 03:00 UTC and keeps recovery points for seven days.



The rule allows one hour to start and two hours to complete. A daily schedule alone does not guarantee a 24-hour recovery point objective.

A scheduled backup completed

The 13 October job protected the original EBS data volume and created the snapshot used in the recovery test.



The console shows the backup plan, completed status and seven-day retention. Console times use UTC+01:00.

The backup record matches

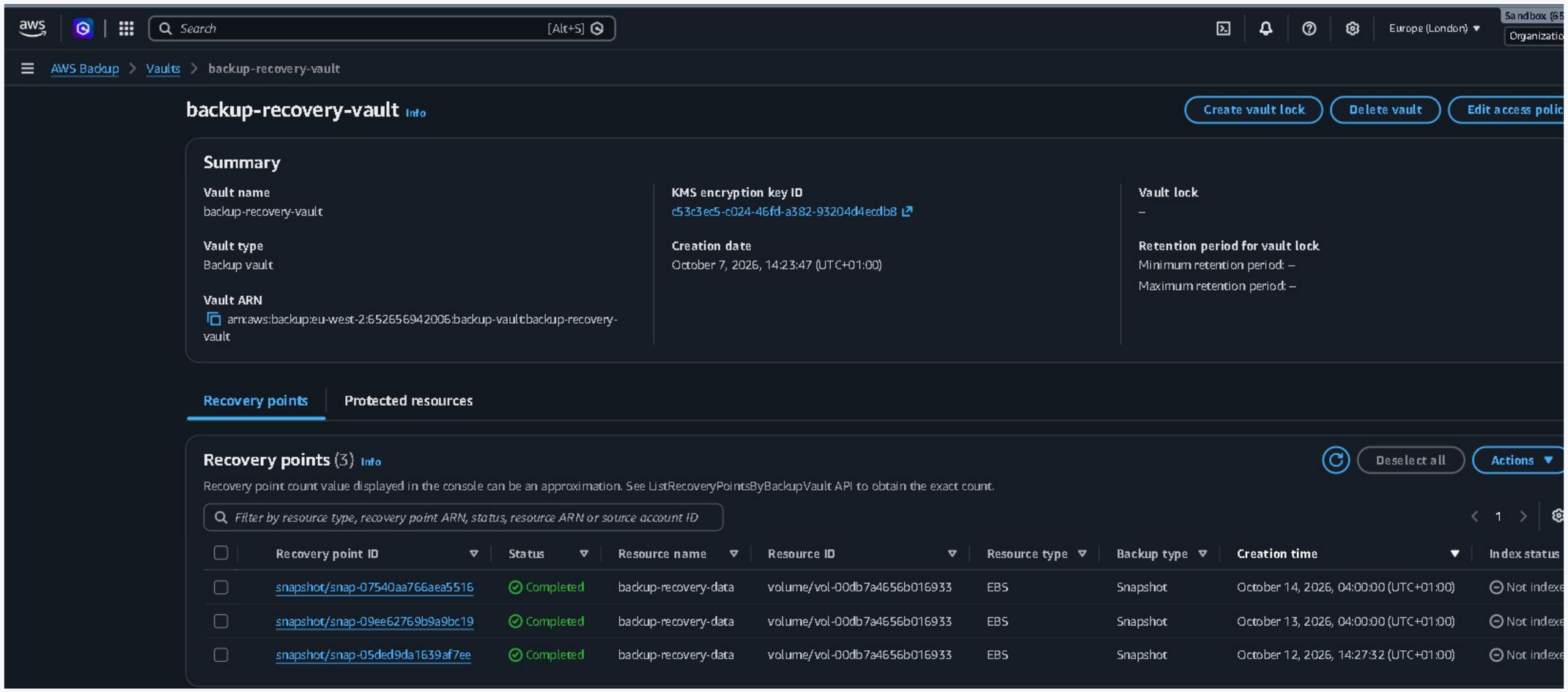
The CLI output records the same scheduled job and snapshot as the console.


```
{
    "BackupJobId": "C2972602-ABB9-AB33-E927-692DC850F626",
    "State": "COMPLETED",
    "Message": null,
    "Created": "2026-10-13T04:00:00+01:00",
    "Completed": "2026-10-13T04:28:38.717000+01:00",
    "Resource": "arn:aws:ec2:eu-west-2:652656942006:volume/vol-00db7a4656b016933",
    "RecoveryPoint": "arn:aws:ec2:eu-west-2::snapshot/snap-09ee62769b9a9bc19",
    "BackupPlanId": "0e10011b-919f-496d-a526-a4707ed045c6"
}
```


Matching the job, source volume and recovery point helps confirm that I am restoring the intended backup.

The vault holds the recovery points

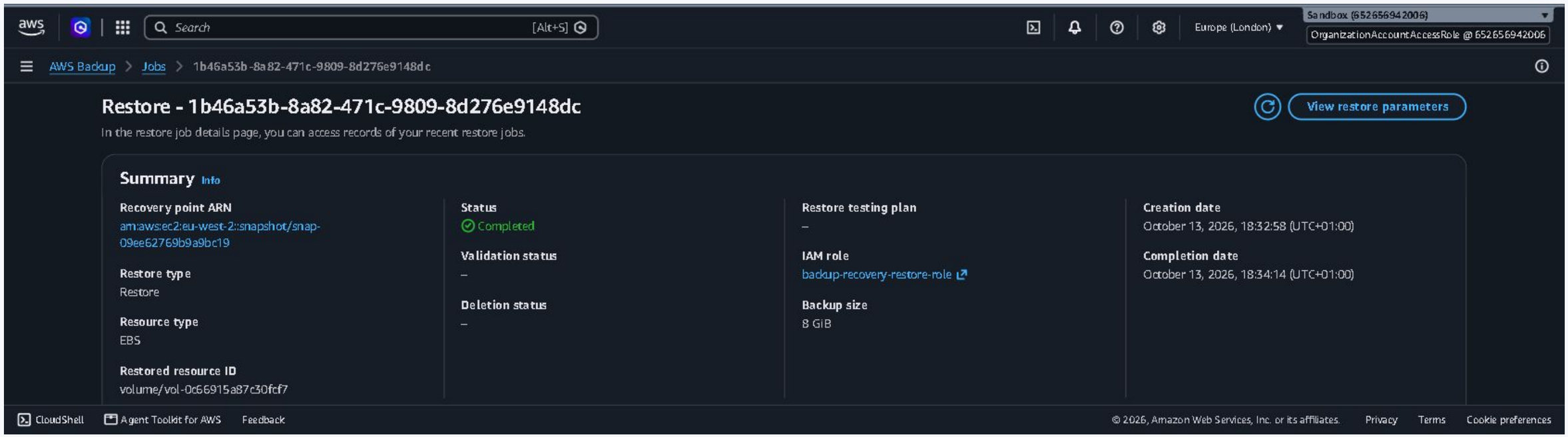
The console shows three completed recovery points, including another scheduled backup on 14 October.



I used the 13 October snapshot for the drill. A newer backup may already contain an accidental deletion, so the recovery point needs checking.

Restoring a separate data disk

After deleting the Keyboard order, I restored the selected snapshot into a new encrypted EBS volume.



AWS completed this storage restore in about 76 seconds. Application recovery also required attaching the disk, starting the server and checking the data.

The restore produced a new volume

The CLI record confirms the completed restore and identifies the new volume.


```
{
    "RestoreJobId": "1b46a53b-8a82-471c-9809-8d276e9148dc",
    "Status": "COMPLETED",
    "Message": null,
    "Created": "2026-10-13T18:32:58.708000+01:00",
    "Completed": "2026-10-13T18:34:14.537000+01:00",
    "RestoredResource": "arn:aws:ec2:eu-west-2:652656942006:volume/vol-0c66915a87c3
0fcf7"
}
```


Terraform then replaced only the recovery server's data-volume attachment. The original data volume remained separate.

The recovery server runs independently

I stopped the original EC2 server and checked that the recovery application still worked.



This later console capture shows the original stopped, recovery running and the temporary setup-test server terminated. The exercise used manual recovery, with no automatic failover.

The recovered orders are readable

```
TARGET SOURCE       FSTYPE OPTIONS
/data  /dev/nvme1n1 xfs    rw,relatime,seclabel,inode64,logbufs=8,logbsize=32k,sunit=8,swidth=8,noquot
NAME          SIZE FSTYPE MOUNTPOINTS SERIAL
nvme0n1         8G                  vol08c12fb1592b3140d
├─nvme0n1p1     8G xfs    /
├─nvme0n1p127   1M
└─nvme0n1p128  10M vfat   /boot/efi
nvme1n1         8G xfs    /data       vol0c66915a87c30fcf7
active
{
  "status": "ok",
  "order_count": 4
}
{
  "orders": [
    {
      "order_id": "ORD-1001",
      "product": "Notebook",
      "quantity": 2,
      "created_at": "2026-10-06T15:44:27+00:00"
    },
    {
      "order_id": "ORD-1002",
      "product": "Keyboard",
      "quantity": 1,
      "created_at": "2026-10-06T15:44:27+00:00"
    },
    {
      "order_id": "ORD-1003",
      "product": "USB cable",
      "quantity": 3,
      "created_at": "2026-10-06T15:44:27+00:00"
    },
    {
      "order_id": "ORD-8692B4044B9E",
      "product": "Monitor stand",
      "quantity": 1,
      "created_at": "2026-10-06T16:02:50+00:00"
    }
  ]
}
2026-10-14T10:19:17Z
```

The application returned all four baseline orders, including the deleted Keyboard order.

This check also shows the restored volume mounted at /data and the service active. The order added after the backup is absent. This screenshot records a follow-up check on 14 October, not the endpoint of the timed drill.

The database passed its checks

SQLite reported no integrity errors, and the file checksum matched the recorded four-order baseline.

```
sh-5.2$ sudo -u orders-app python3 - <<'PY'
import os
import sqlite3

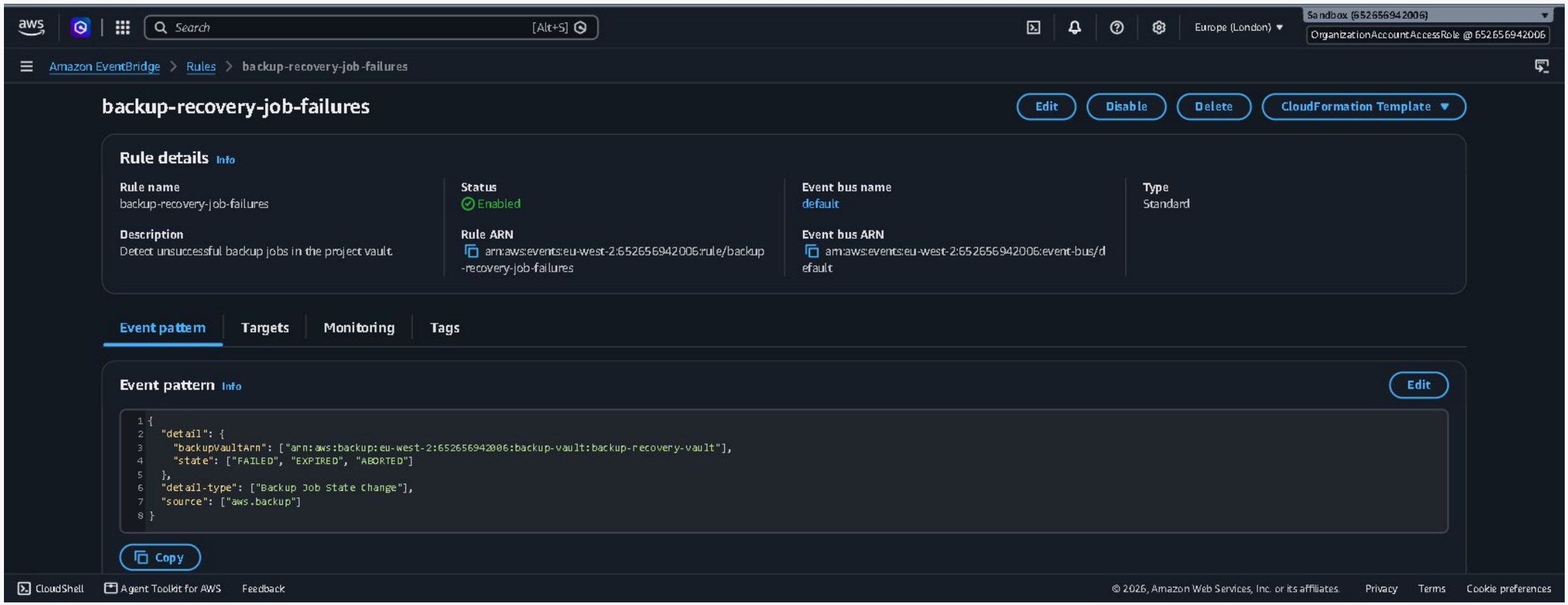
if not os.path.ismount("/data"):
    raise SystemExit("STOP: /data is not mounted.")

connection = sqlite3.connect(
    "file:/data/orders/orders.db?mode=ro", uri=True, timeout=5
)
try:
    results = [row[0] for row in connection.execute("PRAGMA integrity_check")]
    print("Integrity:", results)
    if results != ["ok"]:
        raise SystemExit("STOP: database integrity check failed.")
finally:
    connection.close()
PY
Integrity: ['ok']
sh-5.2$ sudo sha256sum /data/orders/orders.db
date -u +"%Y-%m-%dT%H:%M:%SZ"
4a9f446a0a0b2461917cae20c0392aa6079b93d0722da3290f8c07ead3737d01  /data/orders/orders.db
2026-10-14T10:21:30Z
sh-5.2$
```

I checked both the application response and the database file. This follow-up verification took place at 10:21:30 UTC on 14 October.

Detecting unsuccessful backup jobs

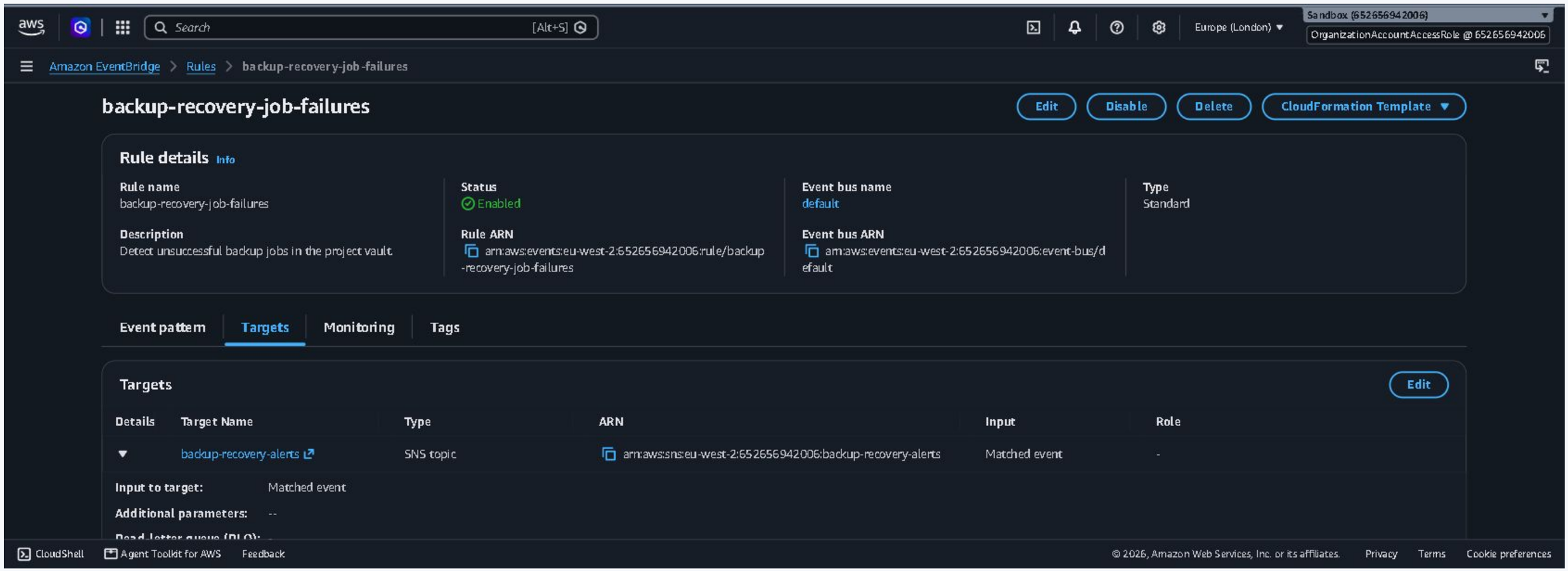
EventBridge watches for FAILED, EXPIRED or ABORTED backup jobs in the project vault.



The rule filters events to this lab. These alerts do not detect every missing or overdue backup.

Routing the event to email

EventBridge sends matching job events to the backup-recovery-alerts SNS topic.



SNS delivers the notification to the confirmed email subscription. The event carries job information, not the database backup itself.

Testing the alert with a cancelled job

I started one extra backup and deliberately cancelled that specific job.


```
{
    "BackupJobId": "505d0ea6-a6f9-400a-a3a9-787f8dfb4472",
    "State": "ABORTED",
    "Message": "Backup job was stopped by user.",
    "Created": "2026-10-13T16:30:25.666000+01:00",
    "Completed": "2026-10-13T16:30:27.561000+01:00",
    "BackupVault": "backup-recovery-vault",
    "Resource": "arn:aws:ec2:eu-west-2:652656942006:volume/vol-00db7a4656b016933"
}
```


AWS reported ABORTED. Later checks found no remaining recovery point or EBS snapshot for this test, so no snapshot cleanup was needed.

The matching email arrived

{"version":"0","id":"02acff8a-f211-08ed-f327-d092b4f6b3f3","detail-type":"Backup Job State Change","source":"aws.backup","account":"652656942006","time":"2026-10-13T15:31:31Z","region":"eu-west-2","resources":[],"detail":{"backupJobId":"505d0ea6-a6f9-400a-a3a9-787f8dfb4472","backupVaultArn":"arn:aws:backup:eu-west-2:652656942006:backup-vault:backup-recovery-vault","backupVaultName":"backup-recovery-vault","bytesTransferred":"0","creationDate":"2026-10-13T15:30:25.666Z","initiationDate":"2026-10-13T15:30:26.047Z","iamRoleArn":"arn:aws:iam::652656942006:role/backup-recovery-backup-role","resourceArn":"arn:aws:ec2:eu-west-2:652656942006:volume/vol-00db7a4656b016933","resourceType":"EBS","state":"ABORTED","statusMessage":"Backup job was stopped by user.","completionDate":"2026-10-13T15:30:27.561Z","startBy":"2026-10-13T23:30:25.666Z","percentDone":0.0,"isParent":false,"resourceName":"backup-recovery-data","vaultType":"BACKUP_VAULT","vaultLockState":"NOT_LOCKED","recoveryPointLifecycle":{"deleteAfterDays":7.0}}}

The email contains the same job ID and ABORTED status as the cancelled backup.

This confirmed the complete AWS Backup, EventBridge, SNS and email path for the cancellation test. I checked the FAILED case with a sample event rather than causing a real failed backup.

What the recovery test proved

8m 5s	Incident to verified application recovery, using an already-prepared recovery server.
4 orders	The baseline orders returned, including the deliberately deleted Keyboard order.
1 later write absent	The post-backup marker did not return. The selected recovery point was about 14 hours 32 minutes old at the incident.
Repeatable setup	I also tested installation scripts on a fresh EC2 server with an existing restored disk, then verified mounting, startup and integrity after reboot.

Scope: one account and one region. No client traffic cutover, live replication or guaranteed recovery time. The main lab still existed when I captured this evidence.